

BPM 1
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Adrenergic Agonists and Antagonists

Theofanis Kollias MD/MPH

Department of Microbiology, Immunology and Pharmacology



St. George's University
Grenada, West Indies

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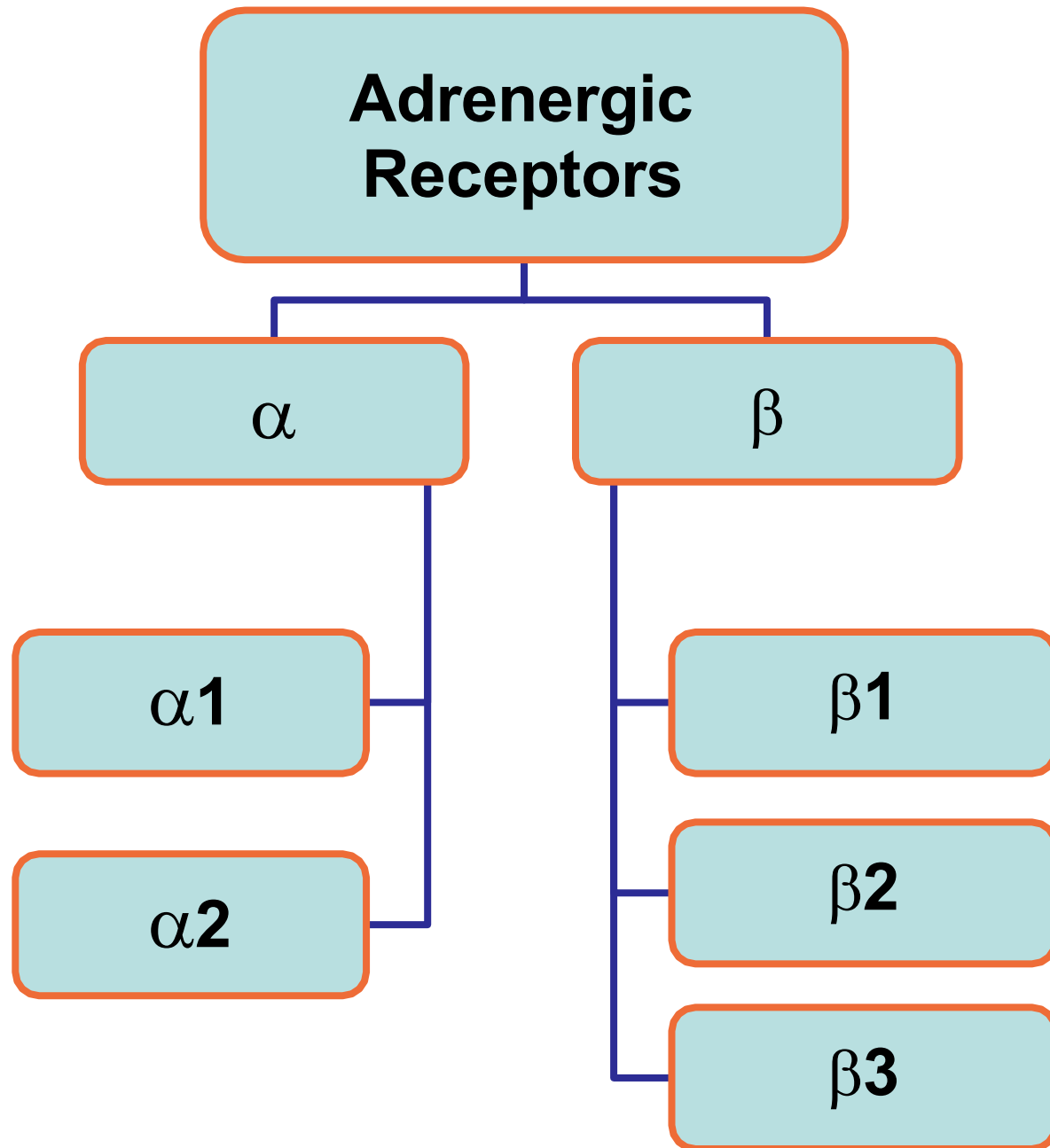
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Objectives

SOM.MK.1.BPM1.1.FTM.4.PHAR.0050	Describe the physiological basis of adrenergic responses.
SOM.MK.1.BPM1.1.FTM.4.PHAR.0051	List the different classes of adrenergic agonists.
SOM.MK.1.BPM1.1.FTM.4.PHAR.0052	Describe the pharmacological effects, uses and adverse effects of the endogenous catecholamines.
SOM.MK.1.BPM1.1.FTM.4.PHAR.0053	Contrast the cardiovascular effects of a low dose and a high dose of epinephrine.
SOM.MK.1.BPM1.1.FTM.4.PHAR.0054	Explain the baroreceptor reflex.
SOM.MK.1.BPM1.1.FTM.4.PHAR.0055	Describe the pharmacological effects, uses and adverse effects of beta agonists.
SOM.MK.1.BPM1.1.FTM.4.PHAR.0056	Describe the pharmacological effects, uses and adverse effects of alpha1 and alpha2 agonists.
SOM.MK.1.BPM1.1.FTM.4.PHAR.0057	Describe the mechanism of action, pharmacological effects, uses and adverse effects of the indirect-acting and mixed-action adrenergic agonists.
SOM.MK.1.BPM1.1.FTM.4.PHAR.0058	List the different classes of adrenergic antagonists.
SOM.MK.1.BPM1.1.FTM.4.PHAR.0059	Describe the pharmacological effects, uses and adverse effects of alpha adrenergic antagonists.
SOM.MK.1.BPM1.1.FTM.4.PHAR.0060	Describe the pharmacological effects, uses and adverse effects of beta adrenergic antagonists. Compare and contrast the adverse effects of the nonspecific and cardiospecific beta adrenergic antagonists.





ADRENERGIC DRUGS

- Adrenergic agonists and antagonists are useful in the treatment of many disorders, such as:
 - Hypertension
 - Angina
 - Heart failure
 - Arrhythmias
 - Asthma
 - Migraine
 - Anaphylactic reactions

ALPHA ADRENERGIC RECEPTORS

REC	SIGNALING	LOCATIONS	EFFECTS
α_1	$\uparrow$ IP ₃ & DAG $\uparrow$ Ca ²⁺ Couples via G _q	Vascular smooth muscle	Contraction
		Genitourinary smooth muscle	Contraction
		Iris radial muscle	Contraction
		Apocrine sweat glands	Secretion
		Liver	$\uparrow$ Glycogenolysis $\uparrow$ Gluconeogenesis
α_2	$\downarrow$ cAMP Couples via G _i	Presynaptic nerve terminals	Inhibition of NE release
		Platelets	Aggregation
		Adipocytes	Inhibition of lipolysis
		Pancreatic β cells	$\downarrow$ Insulin secretion
		Vascular smooth muscle	Contraction (via calcium influx)

BETA ADRENERGIC RECEPTORS

REC	SIGNALING	LOCATIONS	EFFECTS
β_1	$\uparrow$ cAMP Couples via G_s	Heart	$\uparrow$ Heart rate, $\uparrow$ Heart force $\uparrow$ AV conduction velocity
		Juxtaglomerular cells	$\uparrow$ Renin release
		Adipocytes	$\uparrow$ Lipolysis
β_2	$\uparrow$ cAMP Couples via G_s	Smooth muscle (ciliary, bronchial, vascular, gastrointestinal, genitourinary)	Relaxation
		Pancreatic β cells	$\uparrow$ Insulin secretion
		Pancreatic α cells	$\uparrow$ Glucagon secretion
		Liver	$\uparrow$ Glycogenolysis $\uparrow$ Gluconeogenesis
		Adipocytes	$\uparrow$ Lipolysis
β_3	$\uparrow$ cAMP Couples via G_s	Adipocytes	$\uparrow$ Lipolysis

DOPAMINERGIC RECEPTORS

D ₁ , D ₅	↑cAMP Couple via G _s	Smooth muscle of the renal vascular bed	Relaxation
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ADRENERGIC RESPONSES

- The response to adrenergic agonists depends on the proportion of α and β receptors in a tissue.

α Receptors
epinephrine $\geq$ norepinephrine

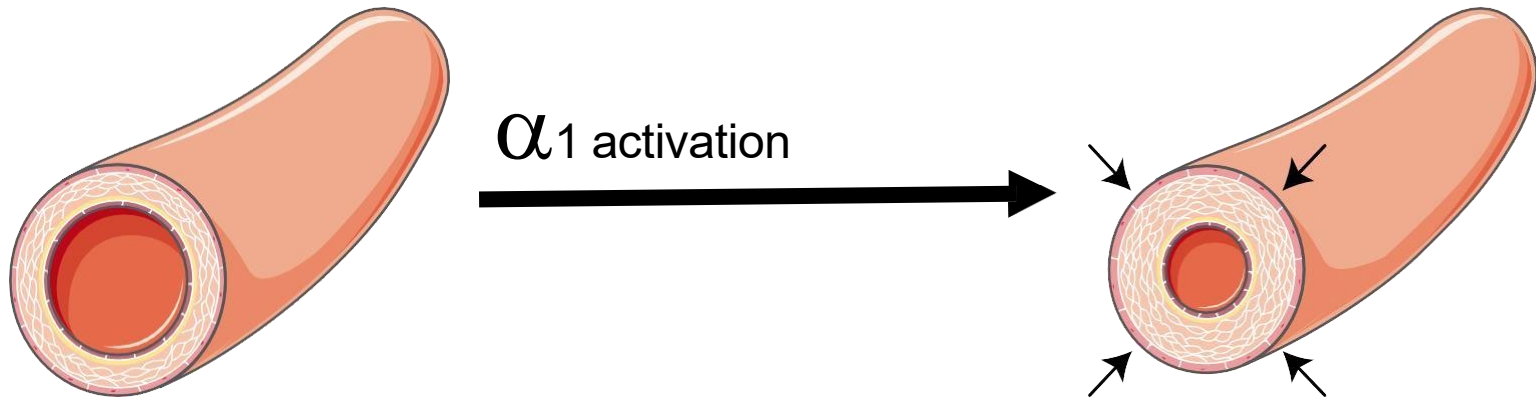
$\beta 1$ & $\beta 3$ Receptors
epinephrine = norepinephrine

$\beta 2$
epinephrine $>$ norepinephrine

- Norepinephrine** has little effect on $\beta 2$ receptors, therefore it cannot cause bronchodilation, because **receptors in bronchial smooth muscle are $\beta 2$** .
- In contrast, **epinephrine** is a potent bronchodilator.

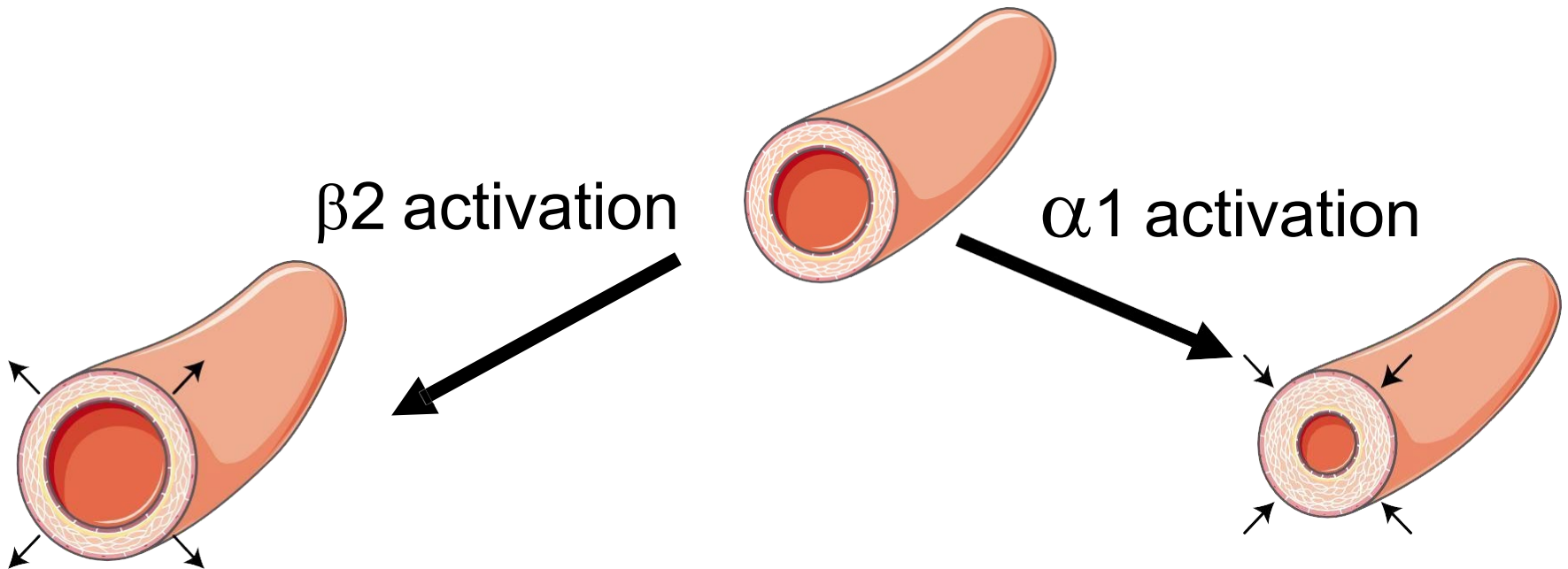
ADRENERGIC RESPONSES

- Skin blood vessels express almost exclusively $\alpha 1$ receptors;
→ thus, norepinephrine and epinephrine cause **constriction of such vessels.**



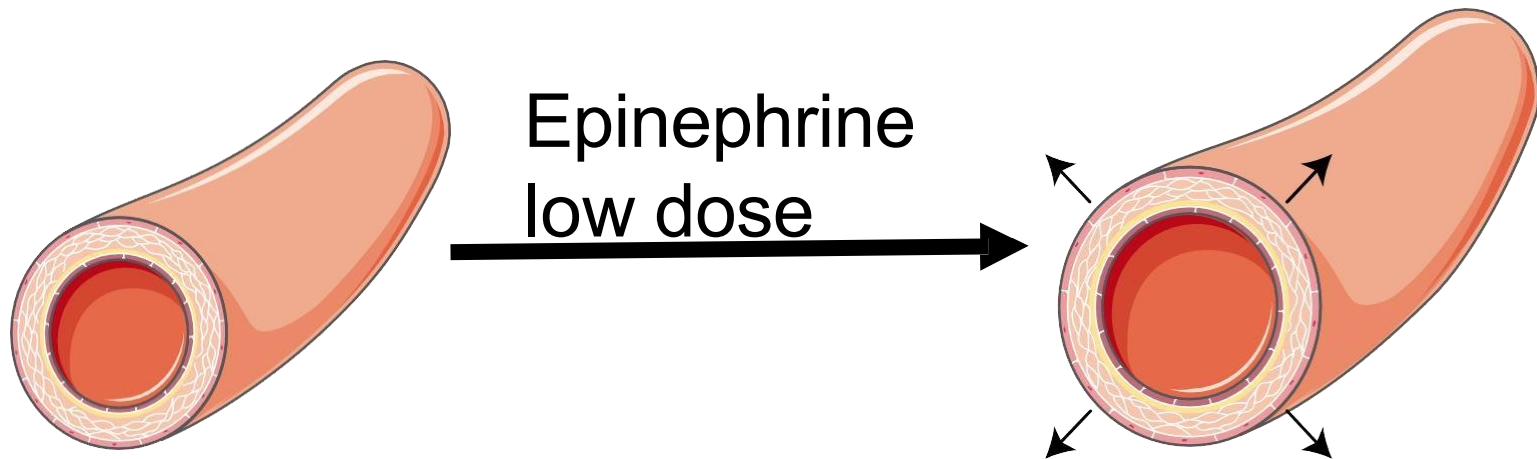
ADRENERGIC RESPONSES

- The smooth muscle of blood vessels in skeletal muscles has both
→ $\alpha 1$ and $\beta 2$ receptors:
- **Activation of $\alpha 1$ receptors → vasoconstriction.**
- **Activation of $\beta 2$ receptors → vasodilation.**



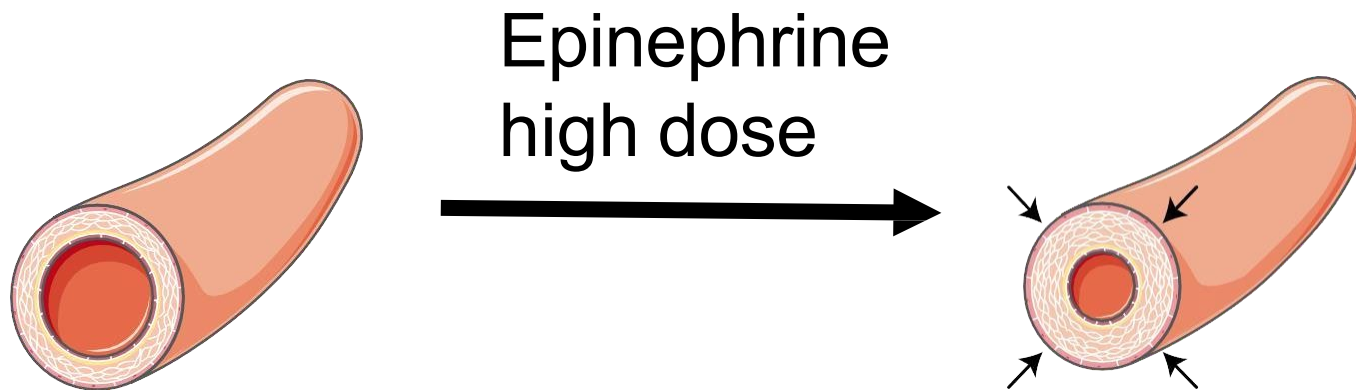
ADRENERGIC RESPONSES

- β_2 receptors are more sensitive to epinephrine than α_1 receptors:
low concentrations of epinephrine cause vasodilation.
- **Physiological concentrations of epinephrine cause primarily vasodilation in such vessels.**



ADRENERGIC RESPONSES

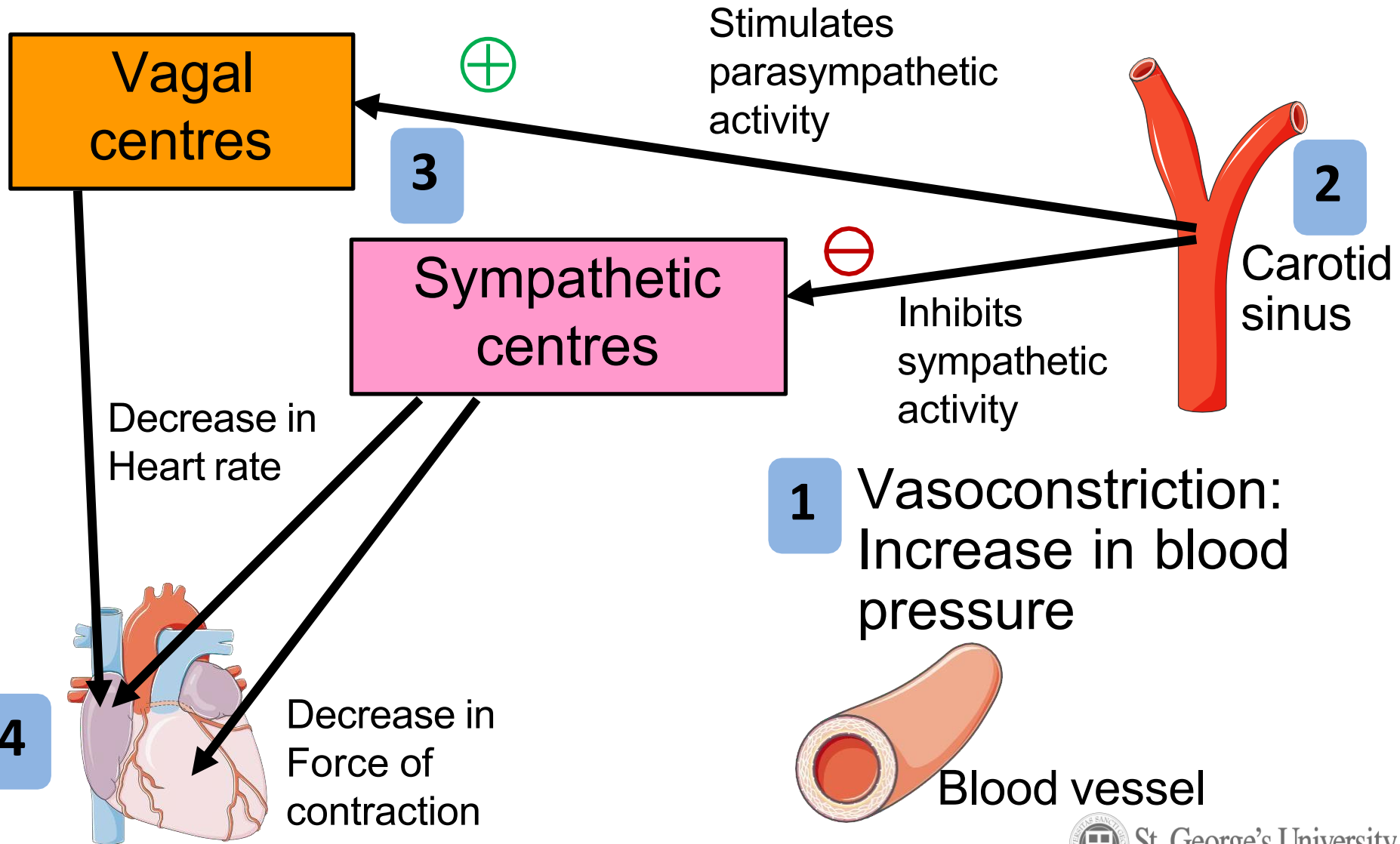
But at **high concentrations of epinephrine**, both $\alpha 1$ and $\beta 2$ receptors are **activated**, and the **response to $\alpha 1$ receptors predominates**, therefore **vasoconstriction results**.



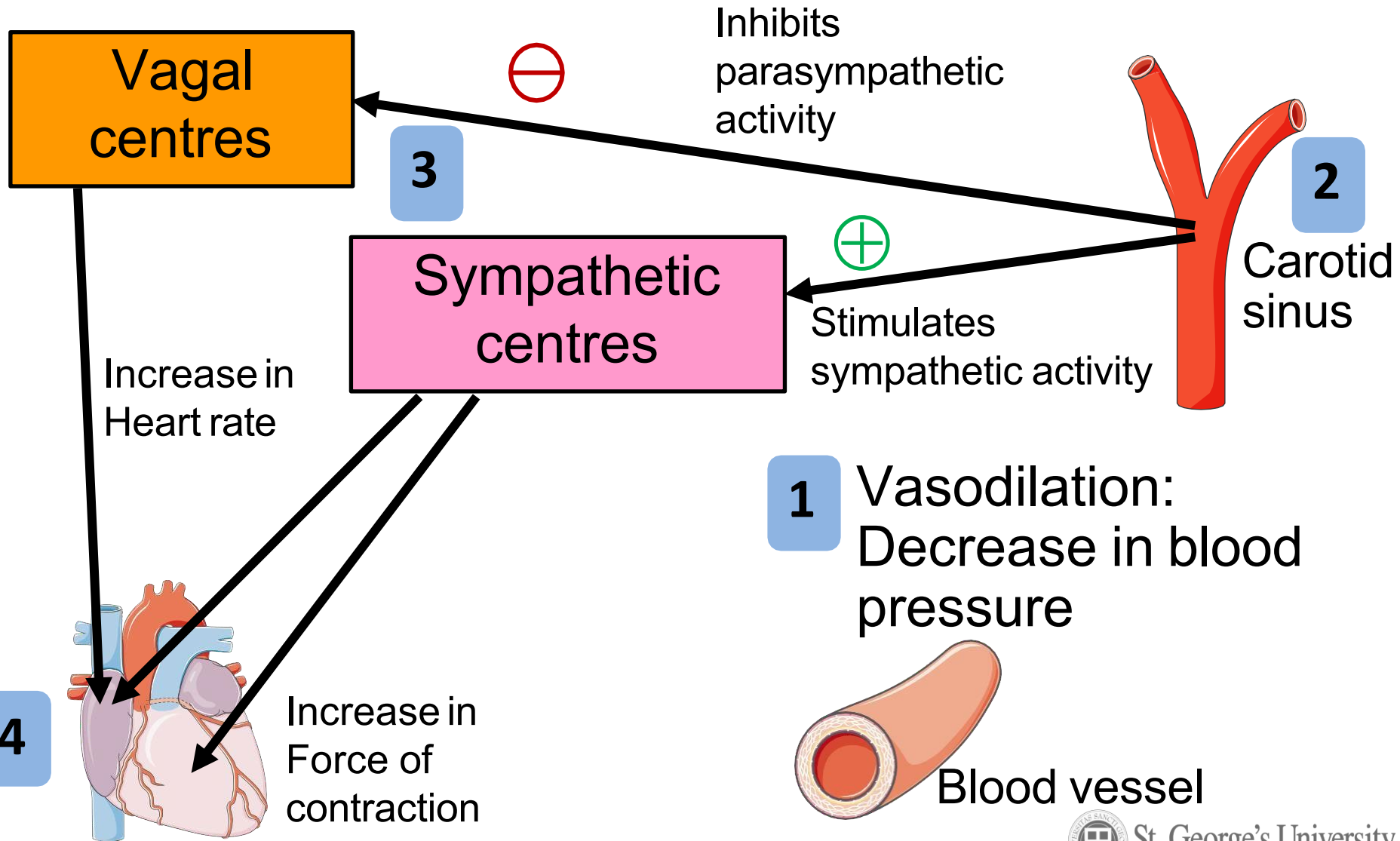
CENTRAL INTEGRATION OF ADRENERGIC RESPONSES

- The ultimate response of a target organ to sympathomimetic amines is dictated **not only by the direct effects** of the agents but also by the **reflex homeostatic responses** of the organism.

THE BARORECEPTOR REFLEX

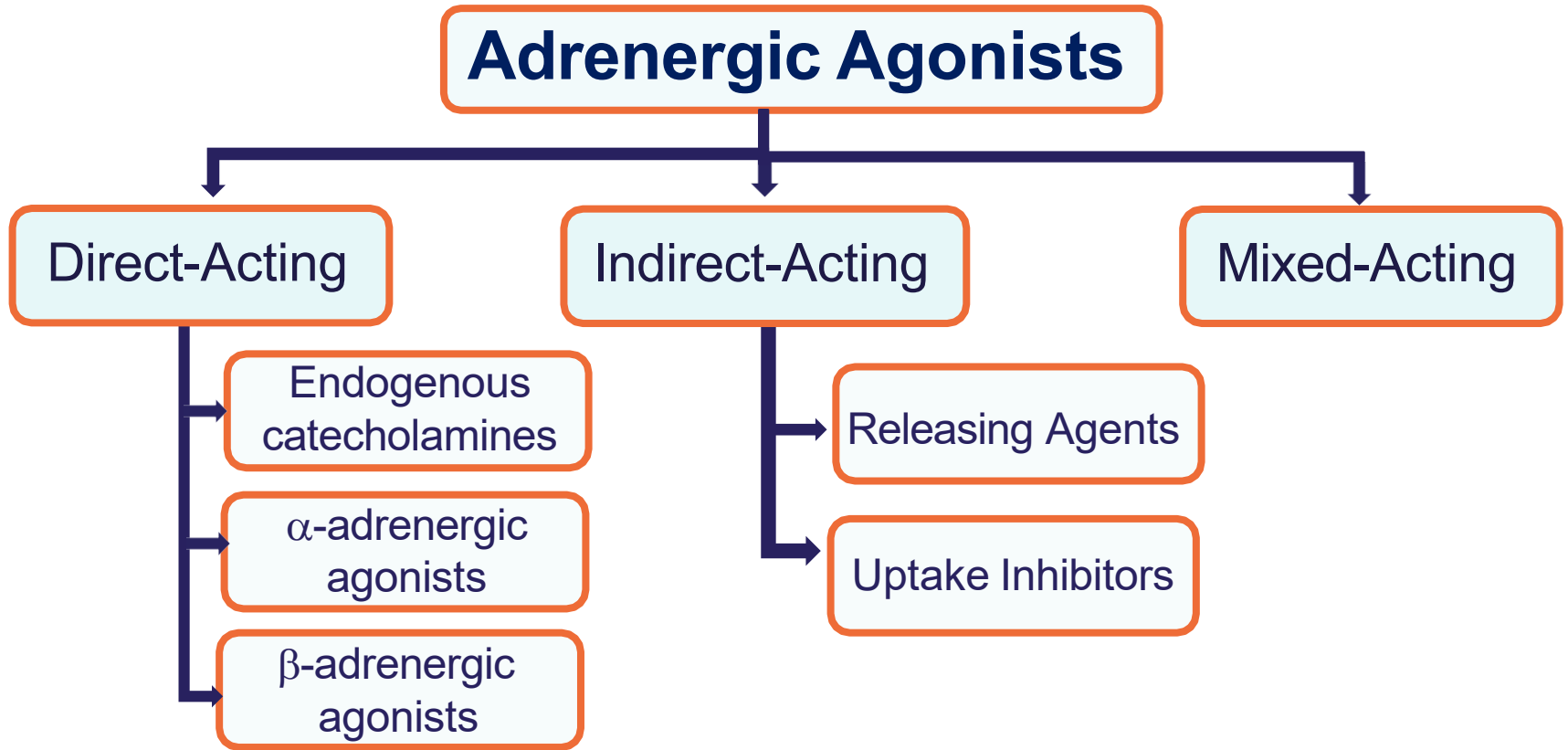


THE BARORECEPTOR REFLEX



ADRENERGIC AGONISTS

Adrenergic Agonists: MOA



DIRECT-ACTING ADRENERGIC AGONISTS

Endogenous Catecholamines

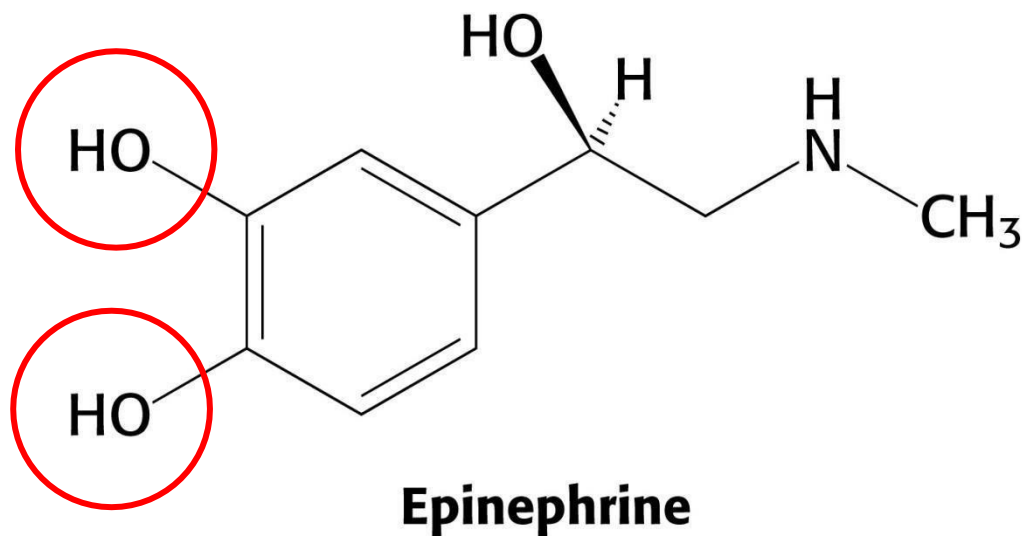
Epinephrine

Norepinephrine

Dopamine

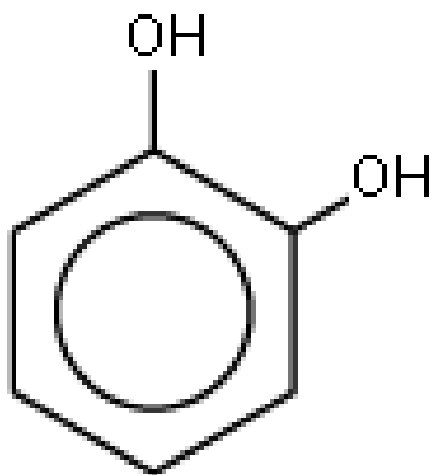
CATECHOLAMINES

- **Norepinephrine, epinephrine, and dopamine** have **–OH** groups substituted at positions **3 and 4** of the benzene ring.

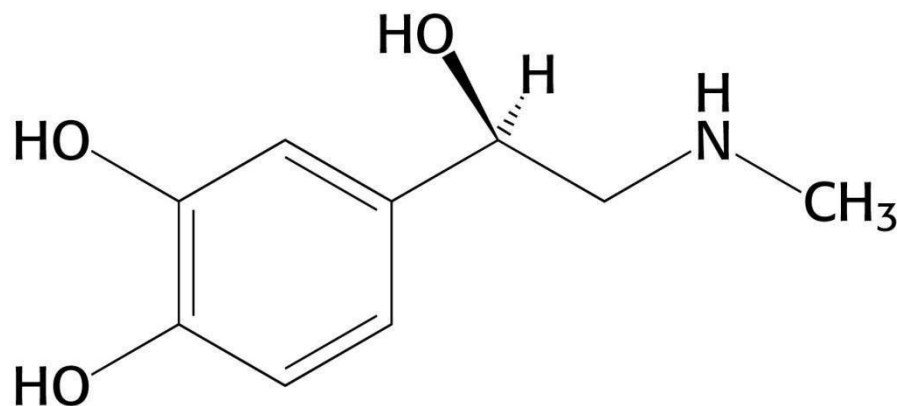


CATECHOLAMINES

- Since **o-dihydroxy-benzene** is also known as **catechol**, sympathomimetic amines with these hydroxyl substitutions in the aromatic ring are called **catecholamines**.



catechol



Epinephrine

Epinephrine

- Epinephrine acts as a **hormone**: after release from the adrenal medulla into the blood it acts on distant cells.
- Epinephrine is an **agonist** at **both** α and β **adrenoceptors**.
- **At low concentrations epinephrine activates mainly β 1 and β 2 receptors.**
- **At higher concentrations α 1 effects become more pronounced.**

Epinephrine: Effects

- Increases heart rate and contractile force (β_1 effect).
- Cardiac output increases: oxygen demand of the myocardium increases.
- Constricts arterioles in skin and viscera (α_1 effect).
- Dilates blood vessels of skeletal muscle (β_2 effect).
- Increases renin release (β_1 effect).
- Relaxes bronchial smooth muscle (β_2 effect).
- Increases liver glycogenolysis (β_2 effect).
- Increases lipolysis (β_1 and β_2 effect).

Effects of Epinephrine on Blood Pressure

When a large dose is given IV

- There is an **increase** in blood pressure due to:

1. α_1 effect – Vasoconstriction

2. β_1 effect

- ✓ Increased ventricular contraction
- ✓ Increased heart rate - This may be opposed by the baroreceptor reflex.

Effects of Epinephrine on Blood Pressure

When a low dose is given IV

1. **$\beta 2$ effect** - **Peripheral resistance decreases**, because $\beta 2$ receptors are more sensitive to epinephrine than $\alpha 1$ receptors → **diastolic pressure falls**.
2. **$\beta 1$ effect**
 - ✓ **Systolic pressure increases** due to increased cardiac contractile force.
 - ✓ **Heart rate increases** - There is no increase in mean blood pressure, so the baroreceptor reflex does not kick in.

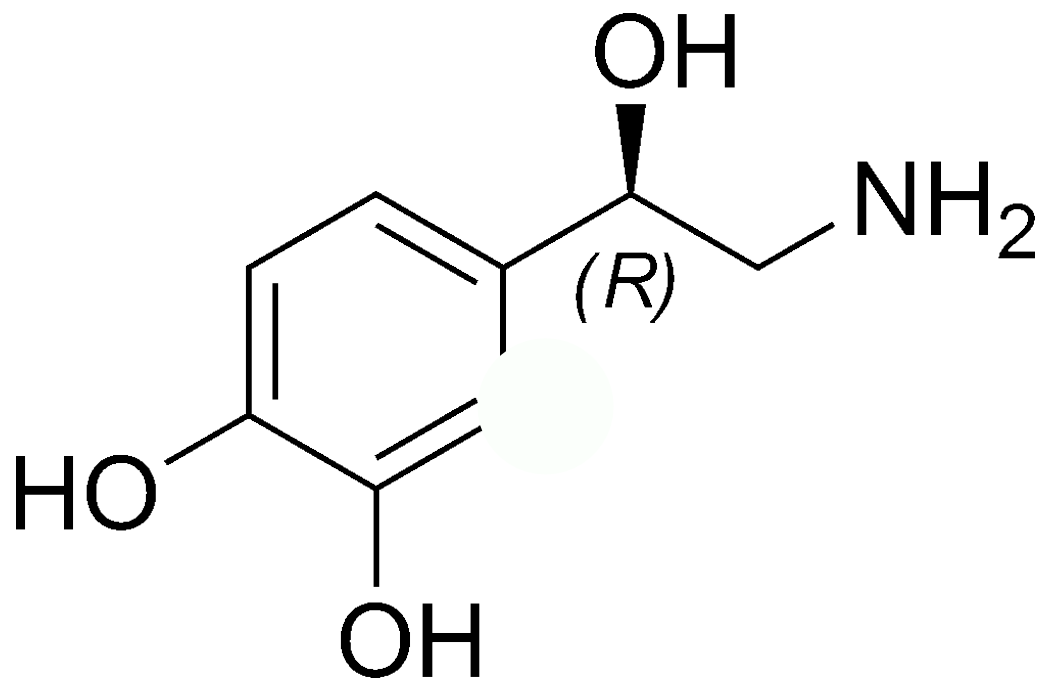
Therefore:
The effects of epinephrine depend on the dose and the ratio of the $\alpha 1$ to $\beta 2$ responses.

Epinephrine: Uses

- **Anaphylactic Shock:** Drug of choice.
- **Acute asthmatic attacks.**
- **Cardiac arrest.**

Norepinephrine

- Agonist at $\alpha 1$, $\alpha 2$, and $\beta 1$ receptors
- Little action on $\beta 2$ receptors



Norepinephrine: Effects

- Causes peripheral vasoconstriction (α_1 effect).
- Increases cardiac contractility (β_1 effect).
- Systemic administration of norepinephrine increases **peripheral vascular resistance**, **systolic** blood pressure and **diastolic** blood pressure.
- **The increase in blood pressure stimulates baroreceptors** leading to reflex bradycardia.

Norepinephrine: Uses

- To treat shock because it increases vascular resistance and therefore **increases blood pressure.**

α -ADRENERGIC AGONISTS

- α 1-SELECTIVE ADRENERGIC AGONISTS
- α 2-SELECTIVE ADRENERGIC AGONISTS

α 1-SELECTIVE ADRENERGIC AGONISTS:

Phenylephrine

- Causes vasoconstriction: α 1 effect.

USES

- Nasal decongestant. Given orally or topically.
- Mydriatic.



α 2-SELECTIVE ADRENERGIC AGONISTS: **Clonidine**

- Partial α 2 agonist.
- Centrally-acting antihypertensive.
- Activates central presynaptic α 2-adrenoceptors.
- Reduces sympathetic outflow. → reduces blood pressure.

β -ADRENERGIC AGONISTS

- **Non-Selective (β_1 & β_2)**
- **β_1 -Selective**
- **β_2 -Selective**



NON-SELECTIVE β -ADRENERGIC AGONISTS:

Isoproterenol

- **Activates β_1 and β_2 adrenergic receptors.**
- **CVS effects**
 - **β_1 effect** → Increases heart rate, force of contraction and cardiac output
 - **β_2 effect** → Dilates arterioles of skeletal muscle → decrease in peripheral vascular resistance.
 - **Diastolic pressure falls.** Systolic blood pressure may remain unchanged or rise.
 - **Mean arterial pressure typically falls.**
- Causes bronchodilation (β_2 effect).
- Isoproterenol may be used in emergencies to stimulate heart rate in patients with bradycardia or heart block.

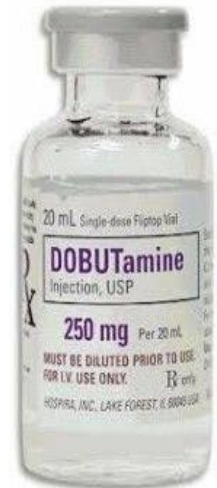
β 1-SELECTIVE ADRENERGIC AGONISTS:

Dobutamine

- Predominantly a β 1 agonist.
- **Potent inotrope, with comparatively mild chronotropic effects.**
- Produces less increase in HR and less decrease in PVR than isoproterenol.
- Increases myocardial O₂ consumption → basis of the **dobutamine stress echocardiogram**.

Uses

- Management of acute heart failure.
- Management of cardiogenic shock.



β 2-ADRENERGIC AGONISTS:

Albuterol

- Causes bronchodilation (β 2 effect).
- **Used with Inhaled Corticosteroids in asthma**
- **DOC for acute asthma attacks**



-ACTING ADRENERGIC AGONISTS

- **Releasing Agents**

- Amphetamine
 - Tyramine
- Cause norepinephrine release from presynaptic terminals.
- Potentiate effects of norepinephrine produced endogenously.

- **Uptake Inhibitors**

- Cocaine

Indirect-Acting Adrenergic Agonists

Releasing Agent: Amphetamine

- Has central stimulatory action.
- Can **increase blood pressure by α -agonist** action on vasculature as well as **β -stimulatory effects on heart.**

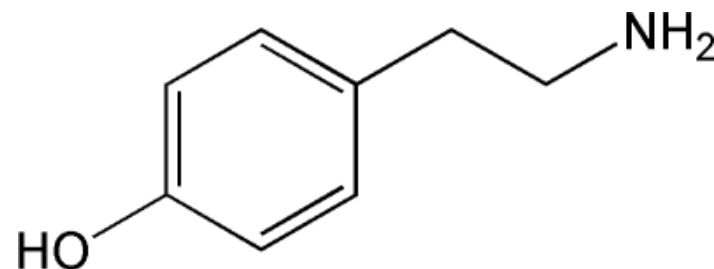
Uses

- ADHD
- Narcolepsy

Indirect-Acting Adrenergic Agonists

Releasing Agent: Tyramine

- Found in fermented foods such as ripe cheese and Chianti wine.
- Normally oxidized by MAO.
- If the patient is taking **MAO inhibitors**, it can precipitate serious vasopressor episodes.
- MAO inhibitors are used as antidepressants.



Indirect-Acting Adrenergic Agonists

Uptake Inhibitors: Cocaine

- Blocks monoamine reuptake.
- Monoamines accumulate in synaptic space.
- This results in potentiation and prolongation of their central and peripheral actions.

MIXED-ACTING ADRENERGIC AGONISTS

Ephedrine, Pseudoephedrine

- Induce release of norepinephrine
- Activate adrenergic receptors.

MIXED-ACTING ADRENERGIC AGONISTS

Ephedrine

- **Not** a catecholamine → poor substrate for COMT → Long duration
- Penetrates the CNS.

USES

- Used as a pressor agent, **particularly** during spinal anesthesia.

MIXED-ACTING ADRENERGIC AGONISTS

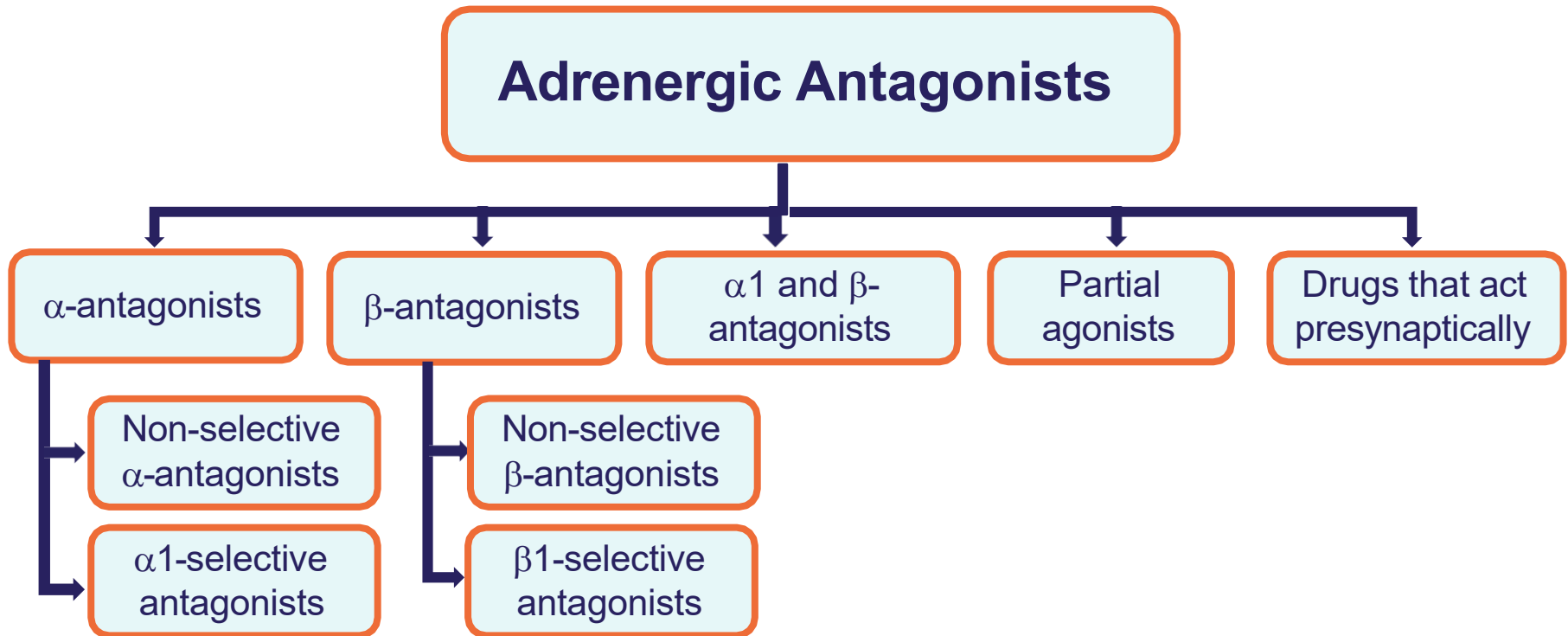
Pseudoephedrine

- One of four ephedrine enantiomers.
- Available over the counter as a component of many decongestant mixtures.



ADRENERGIC ANTAGONISTS

Adrenergic Antagonists: MOA



α -Adrenergic Blockers

- Sympathetic control of the vasculature is mainly due to α 1-adrenergic receptors
- **Blockade** of these receptors *reduces* the sympathetic tone of the blood vessels and *decreases* PVR

Non-Selective α -Adrenergic Blockers

- Phenoxybenzamine
- Phentolamine

α 1-Selective Adrenergic Blockers

- Prazosin
- Terazosin
- Doxazosin
- Tamsulosin

Non-Selective α -Adrenergic Blockers:

Phenoxybenzamine

- **Irreversible** antagonist at $\alpha 1$ and $\alpha 2$ receptors
- Unsuccessful for hypertension

Uses:

- **Pheochromocytoma**
 - Prior to surgical removal of a tumor
 - Chronic management of inoperable tumors

Non-Selective α -Adrenergic Blockers:

Phentolamine

- **Reversibly** blocks $\alpha 1$ and $\alpha 2$ receptors

Uses: • **Pheochromocytoma:** control of hypertension during preoperative preparation and surgical excision

- Diagnosis of pheochromocytoma: phentolamine blocking test
- Prevention of *dermal necrosis* after extravasation of norepinephrine
- Cocaine-induced acute coronary syndrome: to reverse coronary artery vasoconstriction
- **Hypertensive crisis** due to:
 - Stimulant drug overdose
 - Sudden withdrawal of sympatholytic antihypertensive drugs (*eg: clonidine*)
 - Interaction between MAOIs and tyramine or other sympathomimetic amines

α 1-Selective Adrenergic Blockers: Prazosin, Terazosin, Doxazosin, Tamsulosin

Selective blockers of the α 1-receptor

Uses: • Treatment of **hypertension** (*not drugs of choice*)

- Treatment of **Benign Prostatic Hyperplasia (BPH)**
 - Drug of choice for symptom relief
 - Relaxes smooth muscle in the bladder neck, prostate capsule and prostatic urethra, **improving urinary flow**

CV Effects

- Lowers arterial blood pressure by relaxing **both arterial and venous** smooth muscle
- First dose produces an exaggerated hypotensive response that can result in syncope (fainting)
- **The first dose must be 1/4 or 1/3 of normal dose**

α_1 -Selective Adrenergic Blockers

Prazosin (*prototype*) analogs with longer half-life

Terazosin
Doxazosin — Treatment of **hypertension** and **BPH**

- Tamsulosin**
- Three subtypes of the α receptor: $\alpha_1, \alpha_2, \alpha_3$
 - The α_1 receptor predominates in GU smooth muscle
 - **Selective for α_{1A} -receptors**
 - Approved for BPH
 - *Little effect on blood pressure*
 - *Less likely to cause orthostatic hypotension*

β -Adrenergic Blockers

Non-Selective β -Blockers

β_1 -Selective Blockers

α_1 and β -selective blockers

Partial β -Agonists

β -Adrenergic Blockers

Non-Selective β -Blockers

- Propranolol (*prototype*)
- Nadolol
- Timolol

Cardiovascular Effects: β -adrenergic antagonists slow heart rate and decrease myocardial contractility

Respiratory Effects: Blocking β_2 receptors in the lungs can precipitate a respiratory crisis in patients with COPD or asthma

- Nonselective β blockers are **contraindicated** in patients with asthma

Metabolic Effects:
↓ Glycogenolysis
↓ Glucagon secretion

β -Adrenergic Blockers: β 1-Selective Adrenergic Antagonists

Atenolol

Metoprolol

Useful in *hypertensive patients* with impaired pulmonary function

Useful in ***diabetic*** *hypertensive patients* who are receiving insulin or oral hypoglycemic agents

Esmolol

- **Ultra-short** acting
- Half-life ~10 minutes
- IV administration
- Used for rapid control of ventricular rate in patients with atrial fibrillation or atrial flutter

β -Blockers: Uses

Hypertension:	Lowers blood pressure by decreasing cardiac output
Glaucoma:	Effective in diminishing intraocular pressure (<i>particularly Timolol</i>)
Migraine:	Effective for prophylaxis of migraine
Hyperthyroidism:	Blunts sympathetic stimulation
Angina Pectoris:	Decreases O ₂ requirement of heart muscle • Useful in chronic management of stable angina; Not for acute management
Atrial Fibrillation:	Control of ventricular rate

β -Blockers: Uses

Essential Tremor

- Most common neurologic cause of postural or action tremor
- β -Blockers are the most commonly used drugs for treatment

Performance Anxiety ('stage fright'): β -Blockers are the preferred treatment

Myocardial Infarction: Causes a protective effect on the myocardium

β -Blockers: Adverse Effects

Bronchoconstriction

- **Nonselective β -blockers**: Potentially *lethal* side effect in asthmatics
- β 1-selective drugs or drugs with ISA at β 2 receptors may be less likely to evoke bronchospasm

The selectivity of β blockers for β 1 receptors is modest:
They should be avoided *if at all possible* in patients with asthma

Hypoglycemia

- **Nonselective β -blockers** may impair recovery from hypoglycemia in insulin-dependent diabetics due to **blockade of β 2 receptors in the liver**
- $\square\beta$ -blockers mask the tachycardia typically seen with hypoglycemia, denying the patient an important warning sign

A β 1-selective blocker is preferable

β -Blockers: Adverse Effects

Lipid Metabolism

- Blockade of β -receptors *inhibits* release of free fatty acids from adipose tissue
- **Both non-selective and β_1 -selective blockers increase TG and reduce HDL**
- Lipid levels are relatively unaffected by **labetalol** and partial agonists like **pindolol**

CNS Effects

- Sedation
- Dizziness
- Lethargy
- Fatigue

α 1 and β -Selective Blockers: Labetalol, Carvedilol

- Competitive antagonist at β and α 1 receptors
- Used in **hypertension** and **CHF**

Partial β -Agonist:

Pindolol

- Weakly stimulate both beta-1 and beta-2 receptors and have ISA – intrinsic sympathomimetic activity
- May be preferred in individuals with *diminished cardiac reserve* or a propensity to bradycardia

ADRENERGIC DRUGS: DRUG LIST

Adrenergic Agonists

Direct Acting Agonists

Endogenous catecholamines

Epinephrine
Norepinephrine
Dopamine

α 1-agonists

Phenylephrine

β -Agonists

Isoproterenol

α 2-agonists

Clonidine

β 1-Agonists

Dobutamine

β 2-agonists

Albuterol

Indirect Acting

Amphetamine
Tyramine
Cocaine

Mixed Acting

Ephedrine
Pseudoephedrine

ADRENERGIC DRUGS: DRUG LIST

Adrenergic Antagonists

α -antagonists

Phenoxybenzamine

Phentolamine

α_1 -antagonists

Prazosin

Terazosin

Doxazosin

Tamsulosin

β -antagonists

Propranolol

Nadolol

Timolol

β_1 -antagonists

Atenolol

Metoprolol

Esmolol

Partial Agonists

Pindolol

α -1 and β -Selective Blockers

Labetalol

Carvedilol

Common Autonomic Adverse Effects

Muscarinic receptor agonists	Diarrhea, urination, salivation, sweating, miosis, bronchoconstriction, bradycardia, hypotension.
Acetylcholinesterase inhibitors	Diarrhea, urination, salivation, sweating, miosis, bronchoconstriction, bradycardia. Fasciculations, muscle weakness, paralysis. At high doses: hypotension.
Antimuscarinics	Hot, dry skin, xerostomia, mydriasis & cycloplegia, tachycardia, hyperthermia, urinary retention, constipation. Confusion. Delirium.
Amphetamine Cocaine Ephedrine ($\alpha + \beta$ adrenergic)	Hypertension, tachycardia, agitation, mydriasis, diaphoresis.



St. George's University

Grenada, West Indies

Dr. Theofanis Kollias

Department of Microbiology, Immunology and Pharmacology

tkollia1@sgu.edu



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